Project Review:

Next Generation Web: Extended Reality using AI and Splats

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Talk Outline

- Tomorrow's Web
  - Free choice in how you interact and appear in the immersive web
- Technical Challenges
  - Architecture for Immersive Presence
  - Farewell To Triangles
  - Pointillism and Gaussian Splats
  - Hierarchical Reasoning
  - Liquid Neural Networks
  - Pre-training on large corpora
  - Personalization in the browser
- Related standards efforts
- A Web of agents & digital twins
- Democratizing AI development with WebNNM
- Community driven development of the Immersive Web
- Where next for AI and Knowledge-based Systems
- Opportunities for W3C

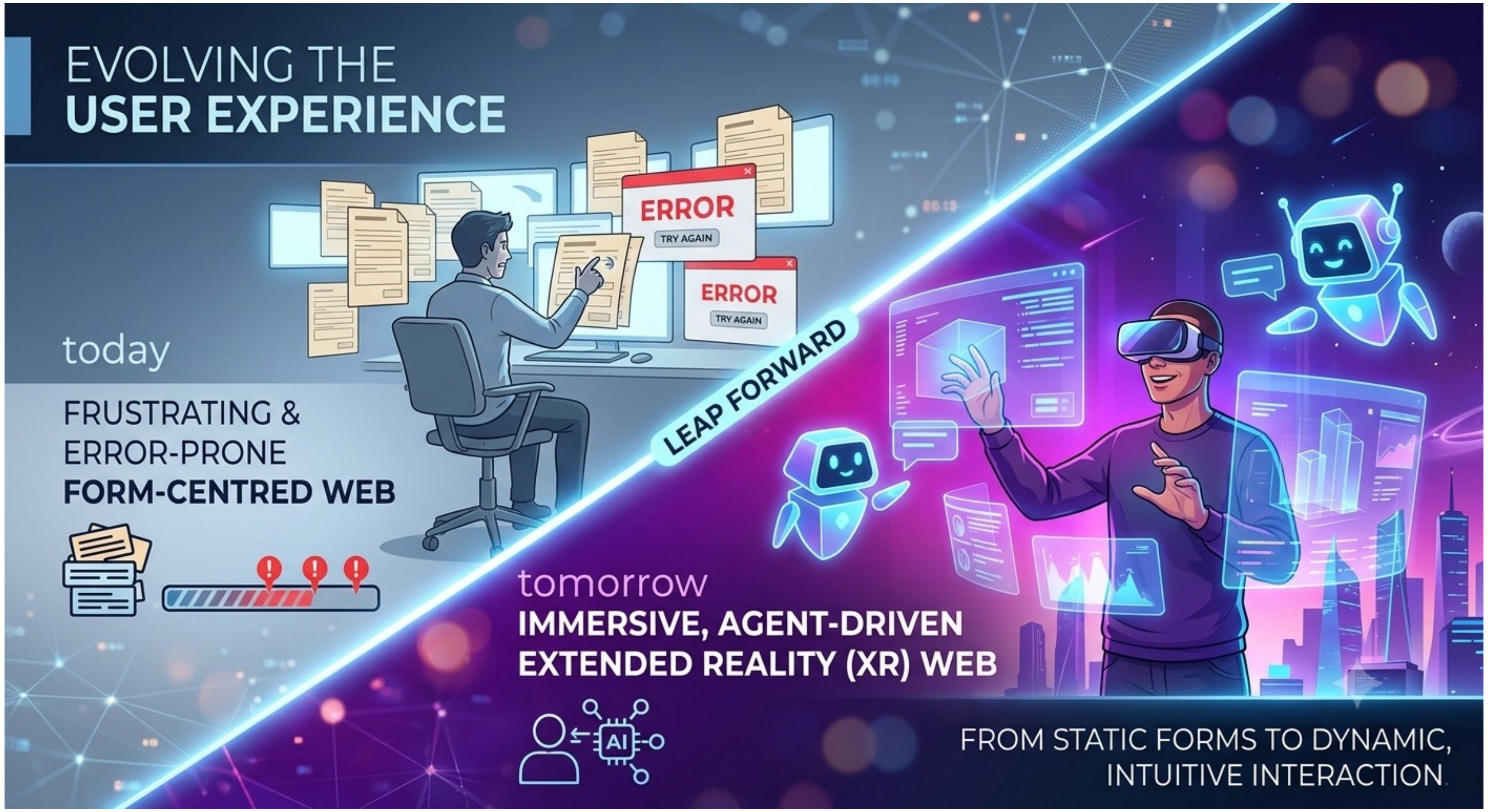
Tomorrow’s Web

Tomorrow’s Web

Today’s Web focuses on form filling which is frustrating and error prone

Partly my fault, as I introduced forms into web pages in the early nineties!

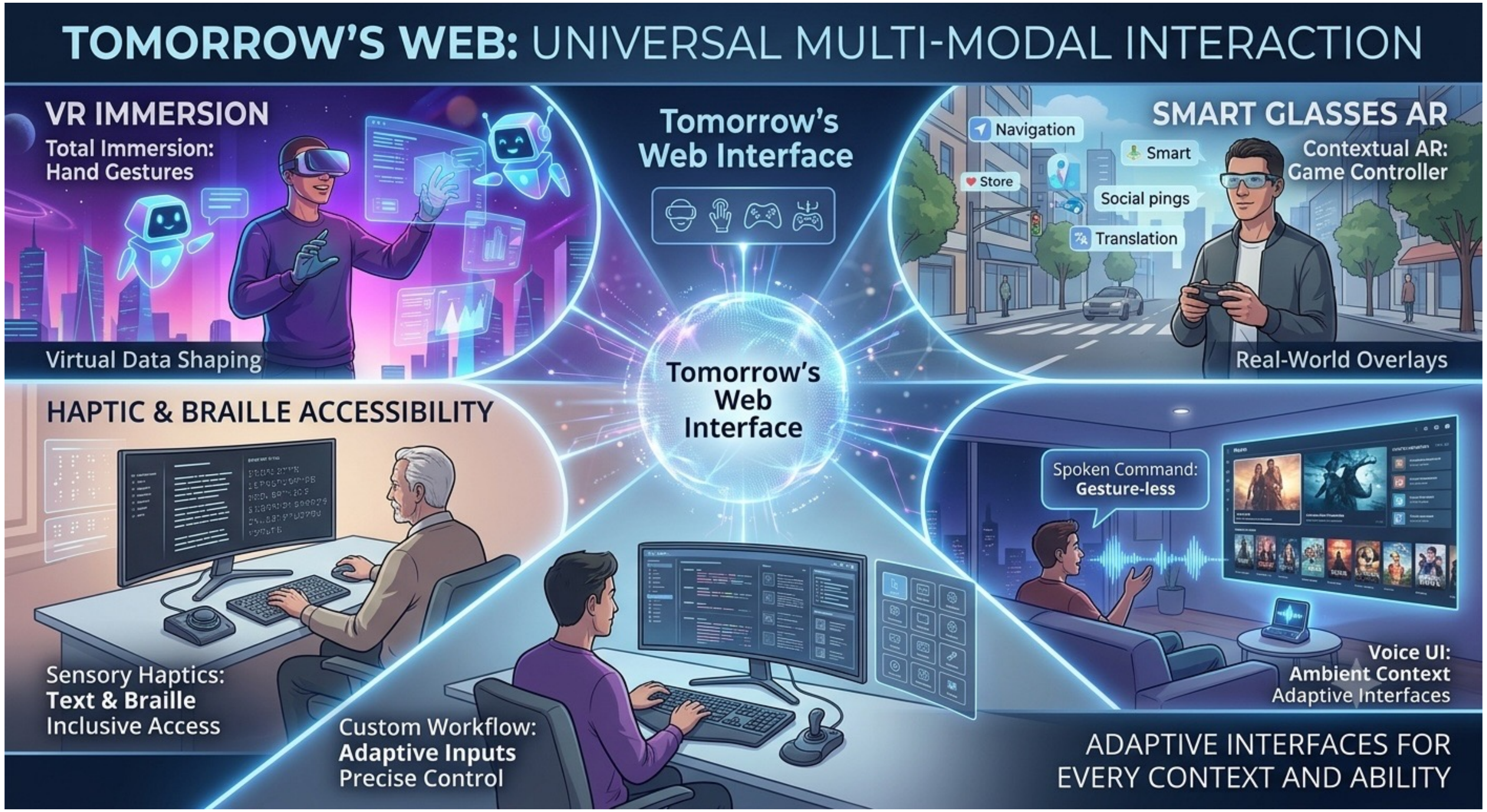
Tomorrow’s Web will instead focus on AI agents and immersive experiences in extended or virtual reality



Personal Freedom in How You Interact

People will be able interact with tomorrow's web in many different ways according to their physical abilities, personal preferences and the social context they find themselves in.

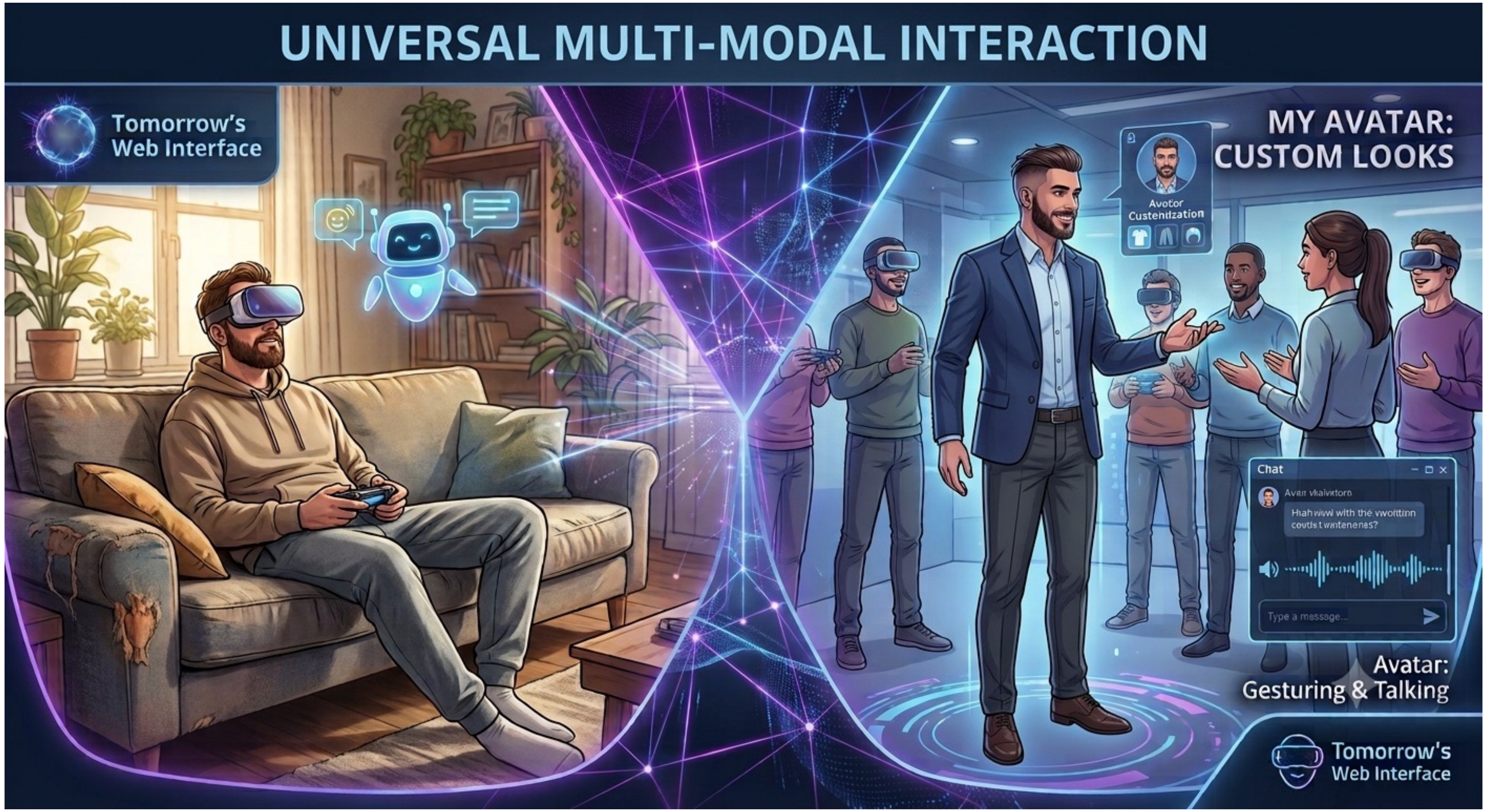
Some people will use VR headsets, others smart glasses, or even regular large monitors, along with games controllers, haptic feedback, regular keyboards, hand gestures and spoken commands.



Personal Freedom in How You Appear

You may be slouched in your living room, but your avatar is walking, talking, gesturing, and looking at people.

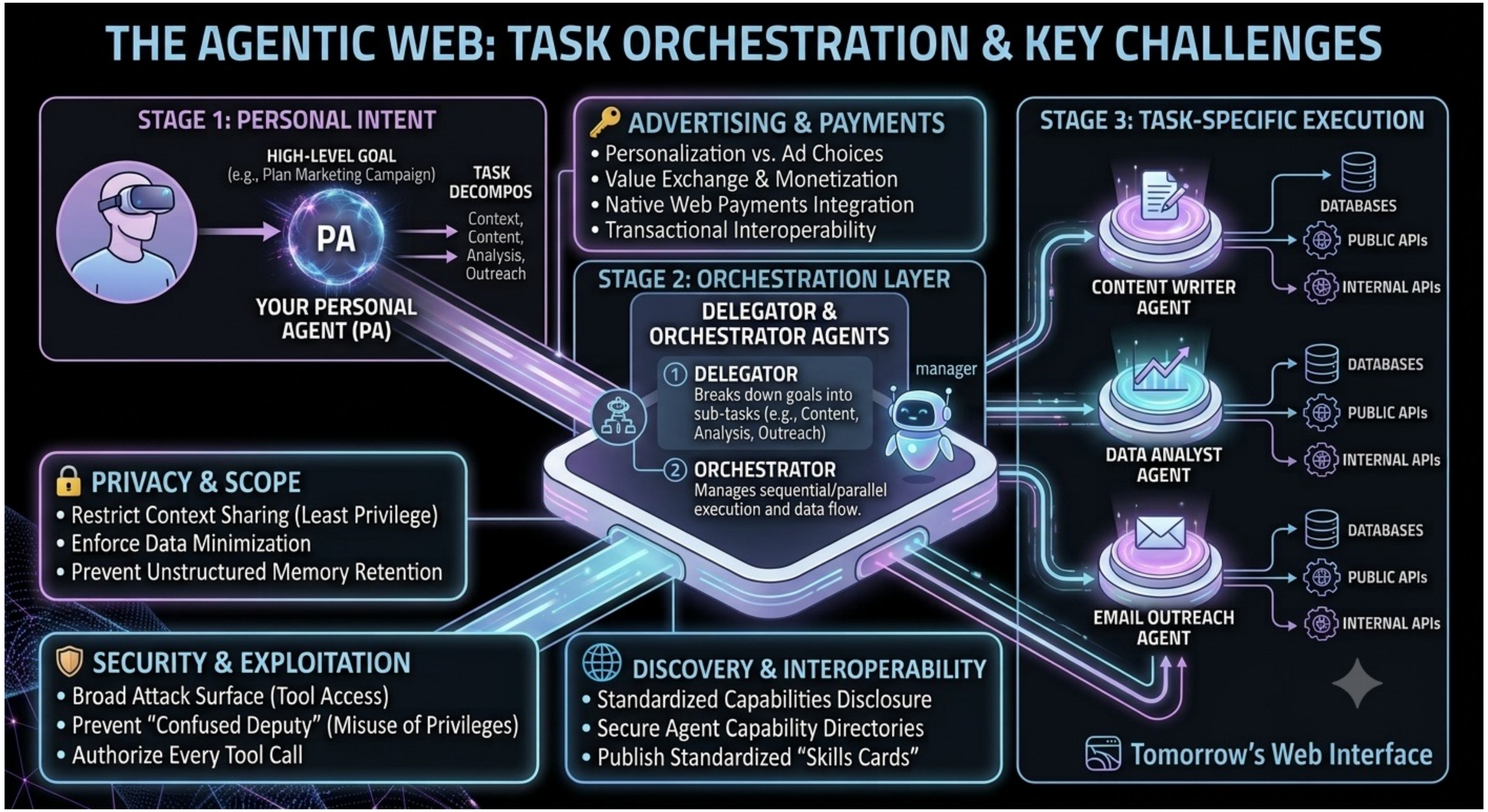
You choose your avatar's appearance to suit the occasion: formal or fun.



A Web of Agents

Your personal agent talks to other agents, which delegate tasks to further agents, which in turn access tools.

Lots of challenges relating to privacy, trust, discovery, advertising, security and payments.

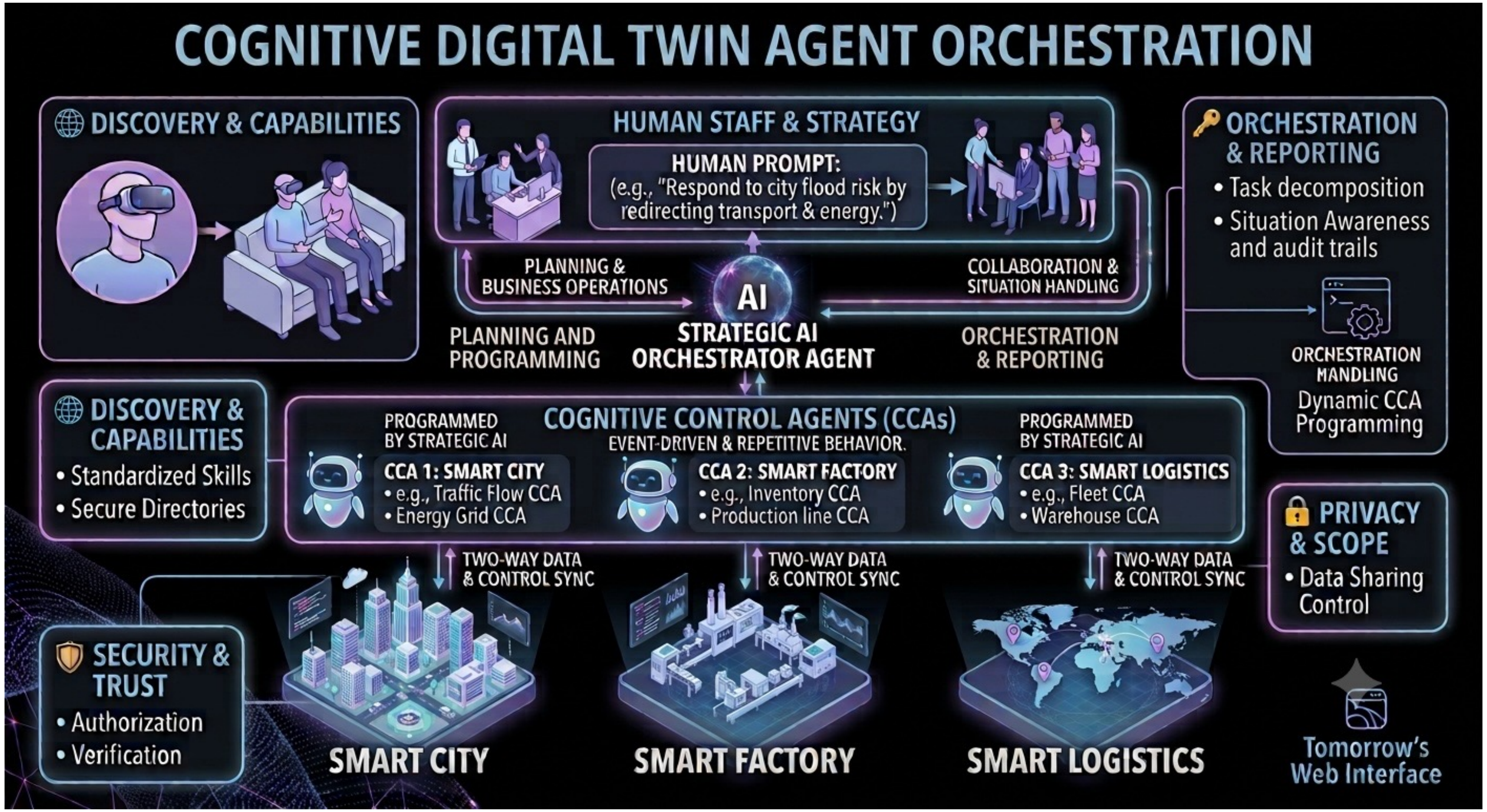


Will advertising to agents replace direct advertising to consumers as agents intermediate services?

Digital Twins and Cognitive Control

A digital twin is a dynamic, real-time virtual model of a physical object, system, or process, using real-world sensors and actuators.

It allows you to simulate performance, predict failures, and test changes in a risk-free digital environment before applying them in reality.



Use of AI agents to program simpler cognitive control agents for event-driven repetitive behaviours

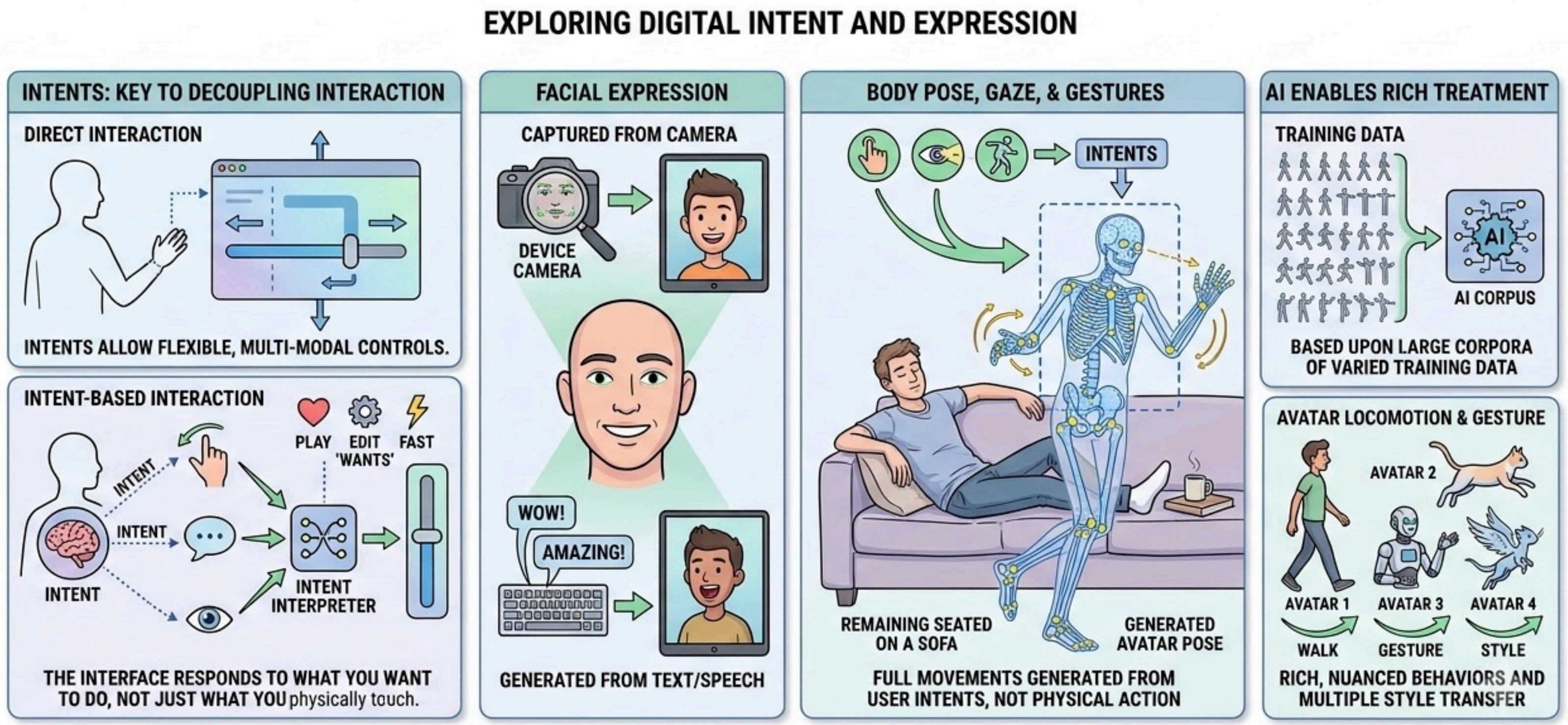
Immersive Scenes and Stage Directions

- In a play, the director interprets the scene descriptions and stage directions, deciding on the concept, mood, era, and visual direction
- Something similar is needed for scenes in virtual reality, where we can exploit generative AI for the role of set and costume designers, starting from a text prompt for the scene description and context
- In place of actors, we have human users and AI agents

[SCENE: *A miniature woodland glade set for a small paper-theater production. The backdrop is a painted watercolor vista of green hills, a forest, and scattered flowers under a cloudy sky. The stage itself is constructed of stained wood, complete with tiny, hand-painted wooden footlights and small red curtains drawn back to reveal the scene. The entire production is rendered in a charming, traditional storybook style.*]
[PRESENT: *AN EXPLORER, A TRAVELER, and A STANDING DEER.*]

Intent-Driven Behavior

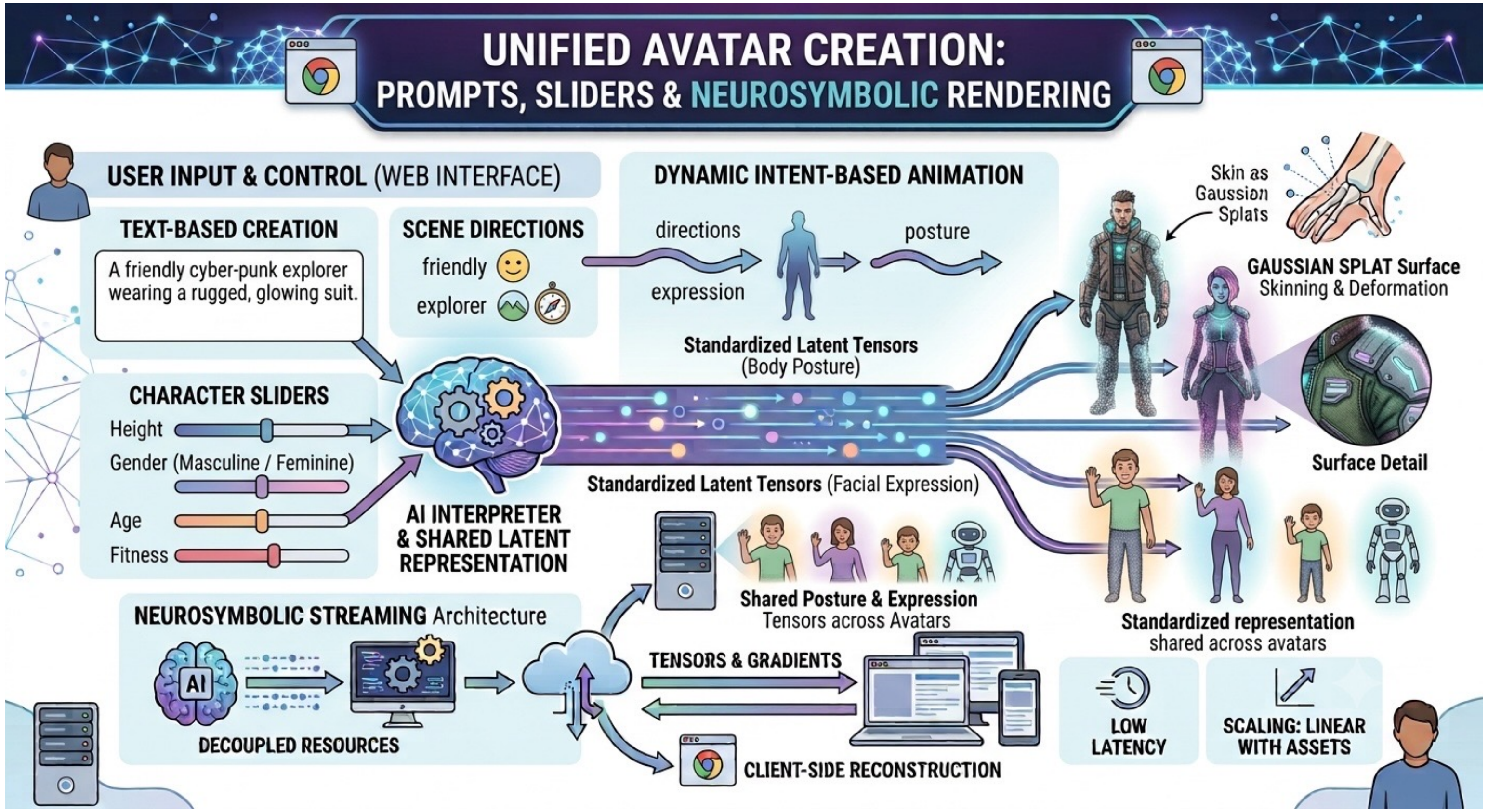
- **Intents** are the key to freeing users to choose how they interact with scenes to direct their chosen avatar's behavior
- Your facial expression may be captured by your device's camera, or it could be generated to match what you are saying or typing
- Body pose, movements, gaze direction and gestures are **intent generated** as you remain seated on the sofa
- AI enables rich treatment of how avatars move* based upon training across large corpora, with individual traits rather than uniformity
- Cognitive mapping from high-level to low-level intents, e.g. "open the door" entails planning a route to the door around any obstacles, moving a hand to the door handle, then opening the door



* Gesturing, standing, sitting, walking, running, dancing, etc.

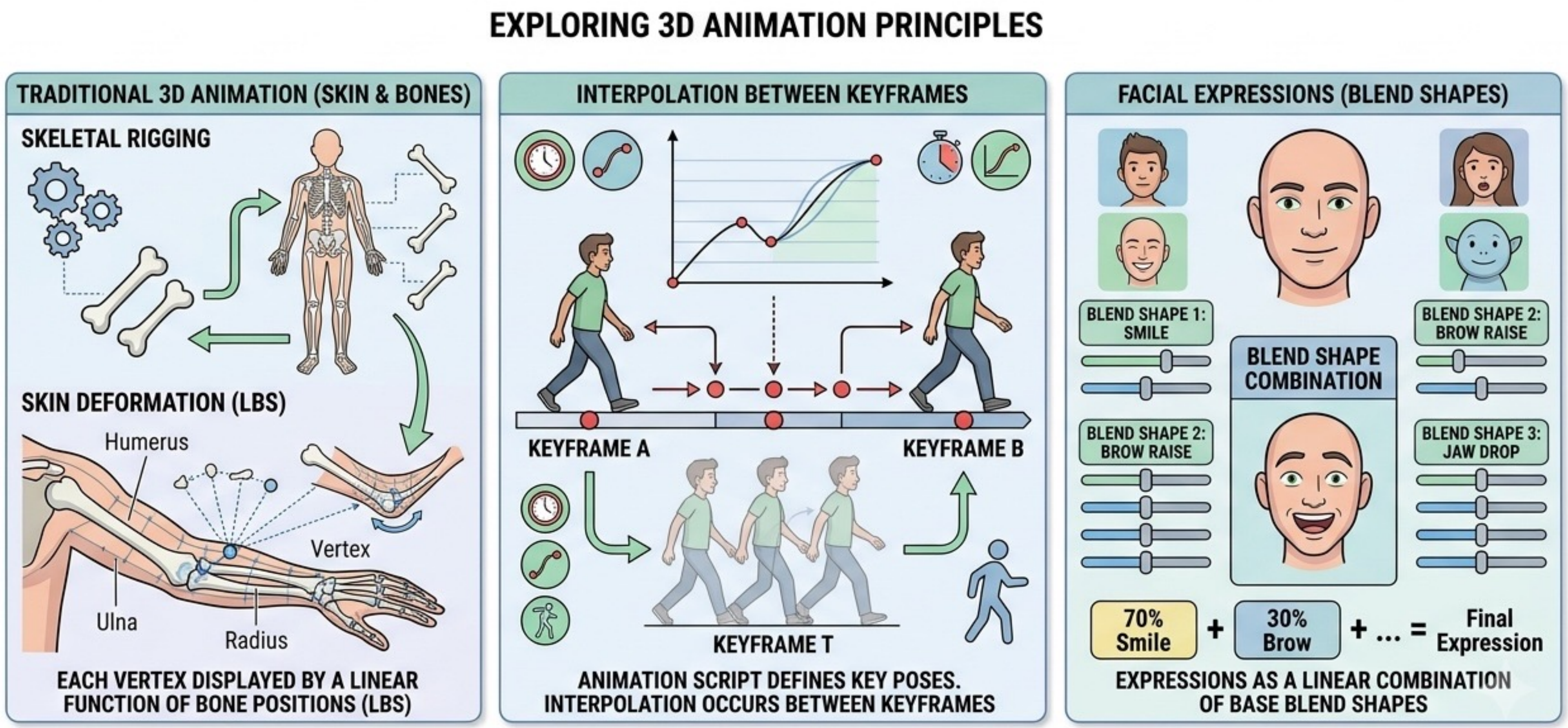
Creating Your Custom Avatar

- Users can simply create their own custom avatars
- Text prompt and sliders, taking scene directions into account
- Avatars have a standardized latent representation for body posture and facial expression
- Key to enabling intent-based animation



Animation using Skin and Bones

- Traditional 3D animates characters in terms of skin and bones
- Each vertex in the mesh forming the skin is displaced by a linear function of the bone positions
- The bone positions are interpolated between key frames in the animation script
- Facial expressions are treated as a linear combination of blend shapes



Technical Challenges

Immersive Web Applications

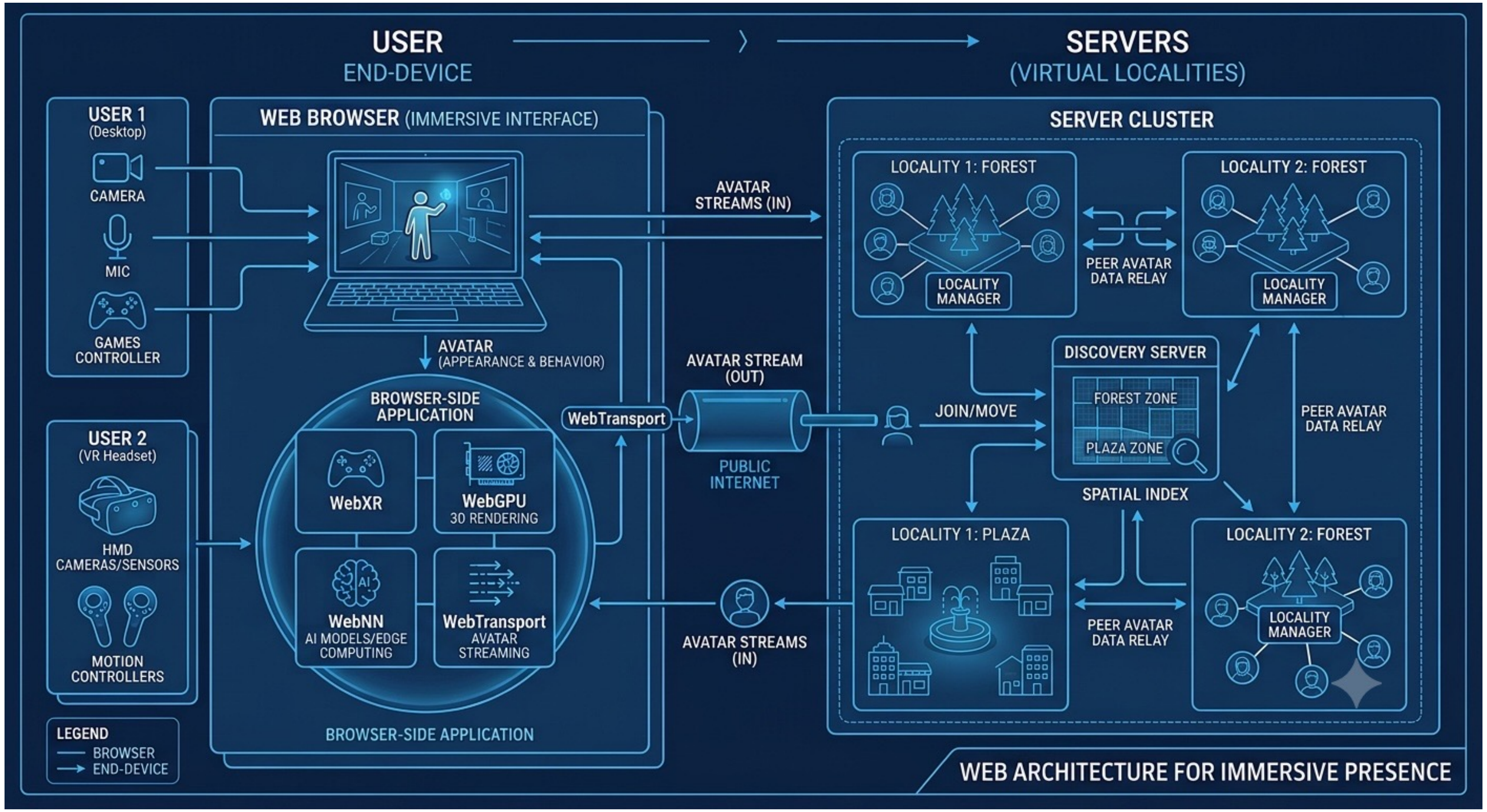
Users access immersive web applications via the browser

WebXR, WebGPU

Browser-side application computes the user's avatar's appearance and behavior, streaming that to other users in the same virtual locality

WebNN, WebTransport

Each server manages one or more localities, along with the means to discover what servers support which localities (spatial indexing), e.g. when joining or moving from one locality to another



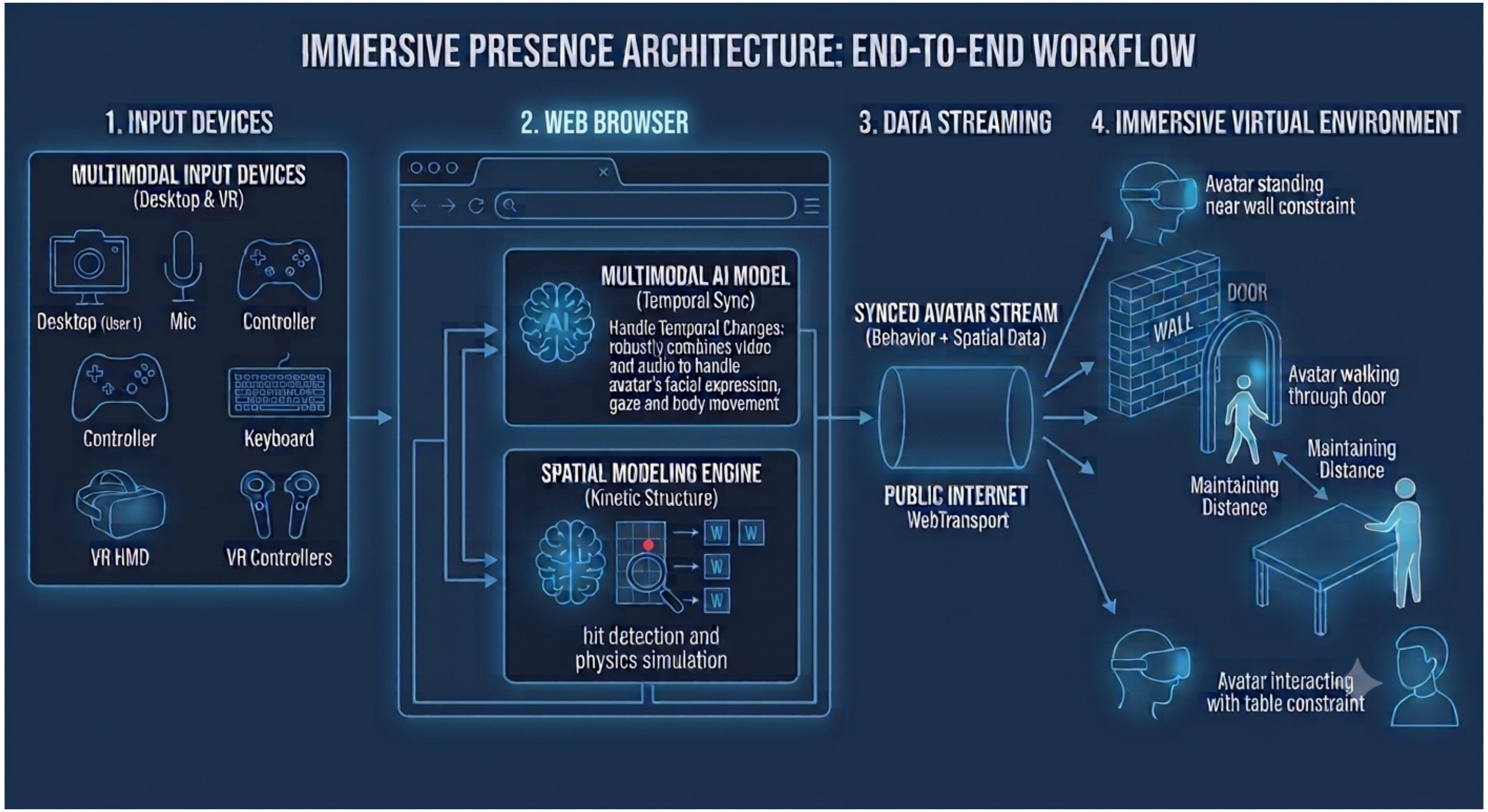
Hit detection and physics simulation are distributed across the browsers to ensure scalable experiences

Exploiting Edge Computing

Multimodal input with camera, mic, games controller, keyboard, …

Multimodal AI model that combines video and audio to robustly handle behaviors as smooth functions of time, including avatar's facial expression, gaze direction and body movement

Spatial modelling to handle everyday physical constraints, e.g. walls, doors, tables, other avatars



Avatars walk on the ground and can't move through solid objects like walls, tables and other avatars

Artificial Neural Networks (ANNs)

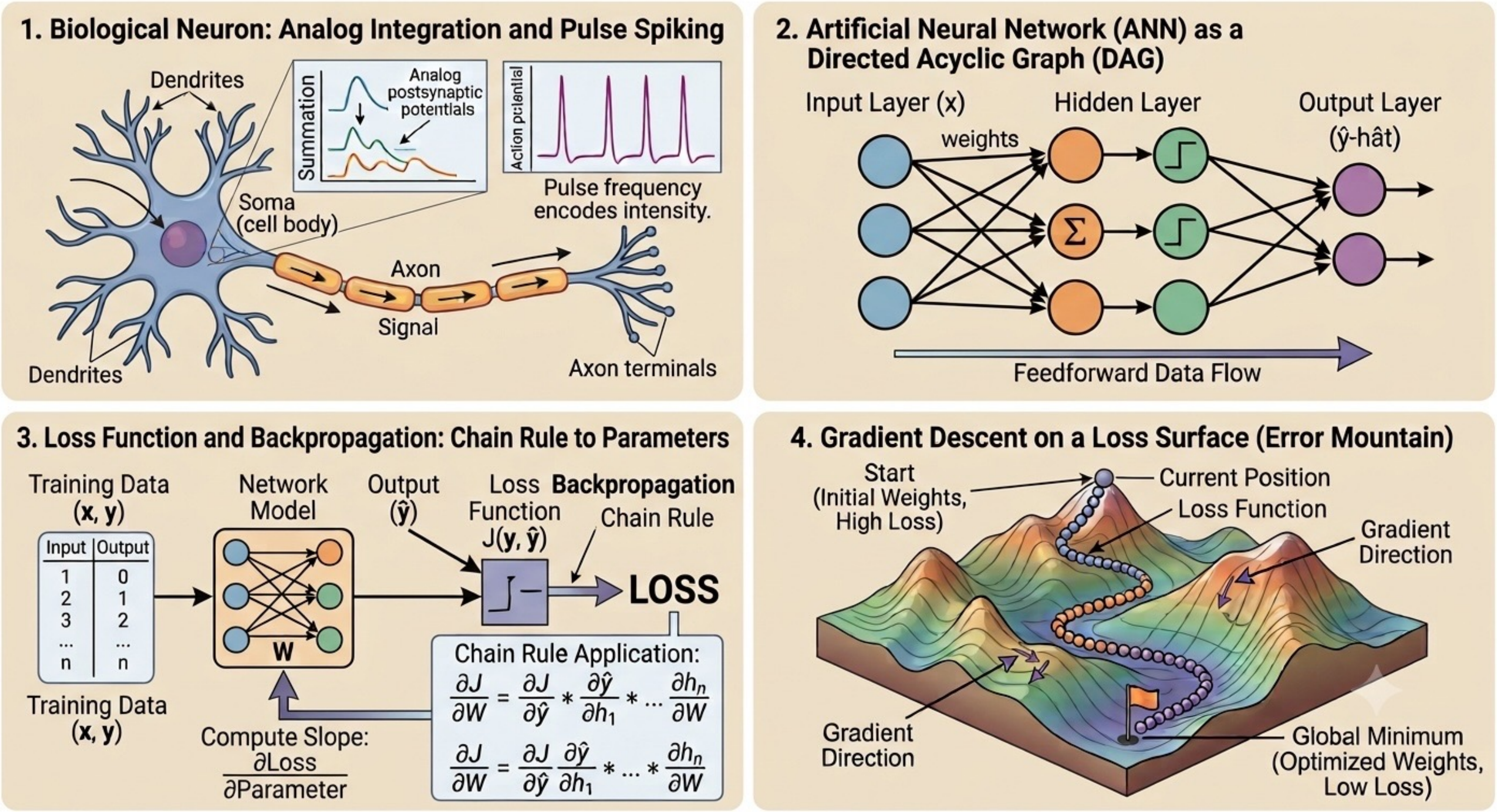
Today's Artificial Neural Networks are different from biological neurons.

ANNs form directed acyclic graphs of mathematical functions acting on tensors (multi-dimensional arrays of numbers).

Training is based upon computing the model's loss given input/output datasets.

Compute gradient of loss with respect to each trainable parameter, by repeatedly applying the chain rule from the outputs until the inputs.

Then adjust each parameter a little in the down-hill direction.



Training is dependent on differentiable loss functions to ensure well behaved gradients

Key Mathematical Operation Groups in ANNs

Operation Group	Purpose	Simple Analogy	Example Math
Linear Transformations	How we apply the model's knowledge (parameters) to the data.	**Filtering/Dials:** Turning up the importance of specific features.	$y = (\mathbf{Weight} \cdot x) + \mathbf{Bias}$
Non-linear activations	How we introduce complex decision-making logic, preventing the model from just being a chain of simple averages.	**The "Trigger" Logic:** A threshold that decides if a neuron is "active" (e.g., "only care if the number is positive").	ReLU (Rectified Linear Unit): **max(0, x)**
Convolution/Pooling	Essential for images/patterns. Convolutions highlight features; Pooling reduces data size while keeping the most important info.	**Scanning & Downsizing:** Highlighting edges, then making a simplified thumbnail.	2D Matrix Convolution; Max-Pooling
Normalization	Keeps the numbers in a healthy range, preventing them from exploding or vanishing, which makes learning more stable.	**Standardizing:** Ensuring every input ingredient is measured on the same scale (0 to 1).	Batch Normalization
Loss Function	How the model measures its mistake. This is the height of "Error Mountain."	**The Score:** The absolute difference between the predicted answer and the true answer.	Mean Squared Error (MSE)

GPUs speed repeated operations over large arrays of numbers

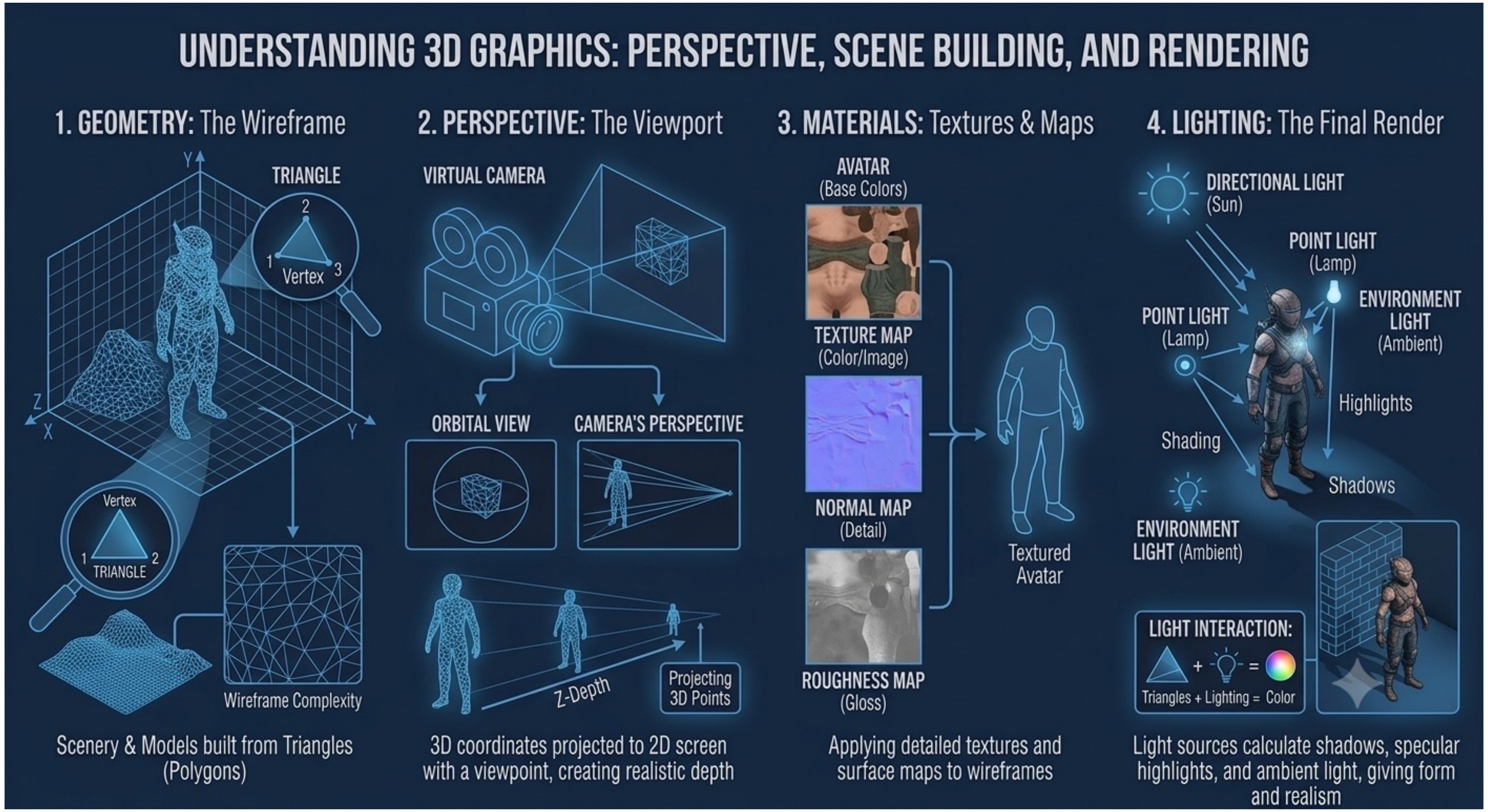
3D Graphics and Problems with Triangles

3D Graphics are based upon, **triangles**, **textures**, **lighting** and **perspective** with lots of matrix math.

WebGPU Shaders are programs you upload to GPUs for higher speed execution.

But triangles have abrupt edges and as such are not differentiable.

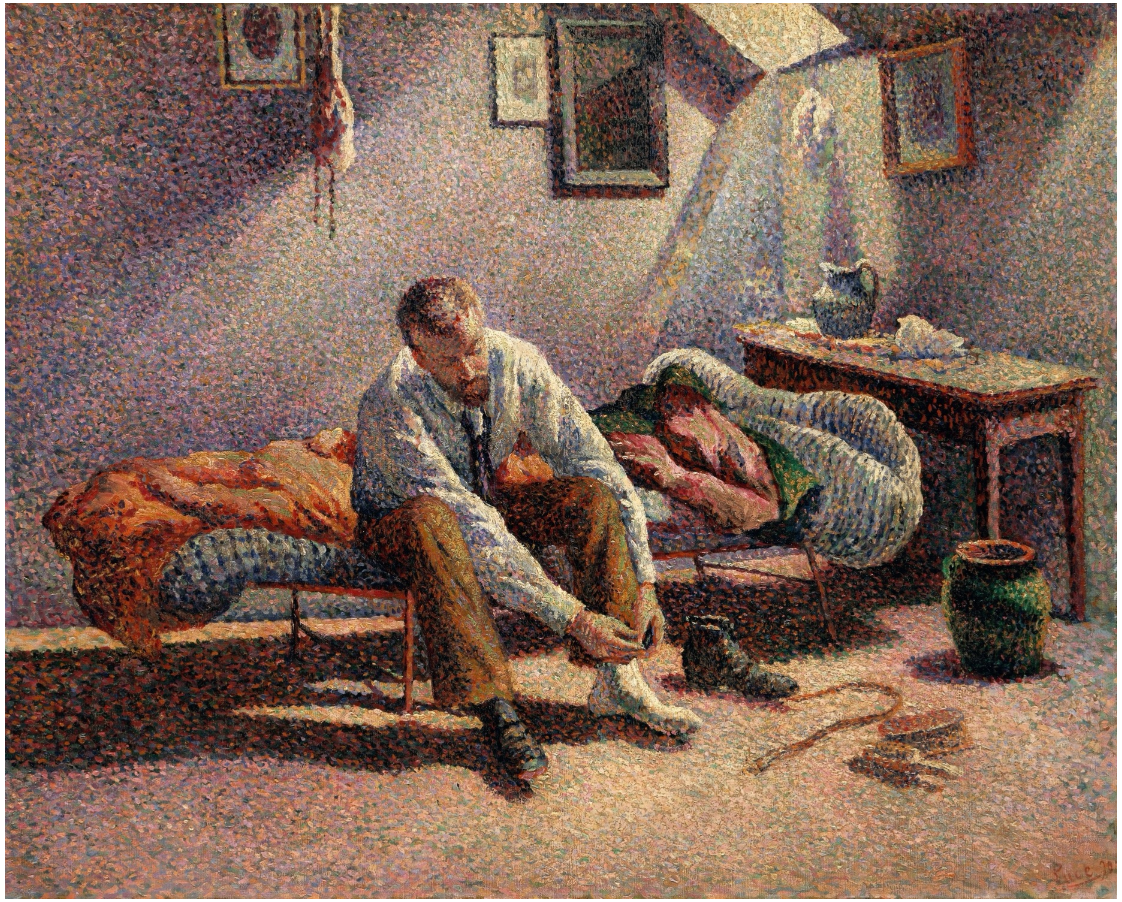
This makes triangles a **poor fit** for use with AI models.



Potential work arounds include triangles with fuzzy edges or using 3D Gaussians

Pointillism as part of Impressionism

Instead of mixing colors directly, use distinct blobs of colors leaving the eye to blend them together: Georges Seurat and Paul Signac (1886)



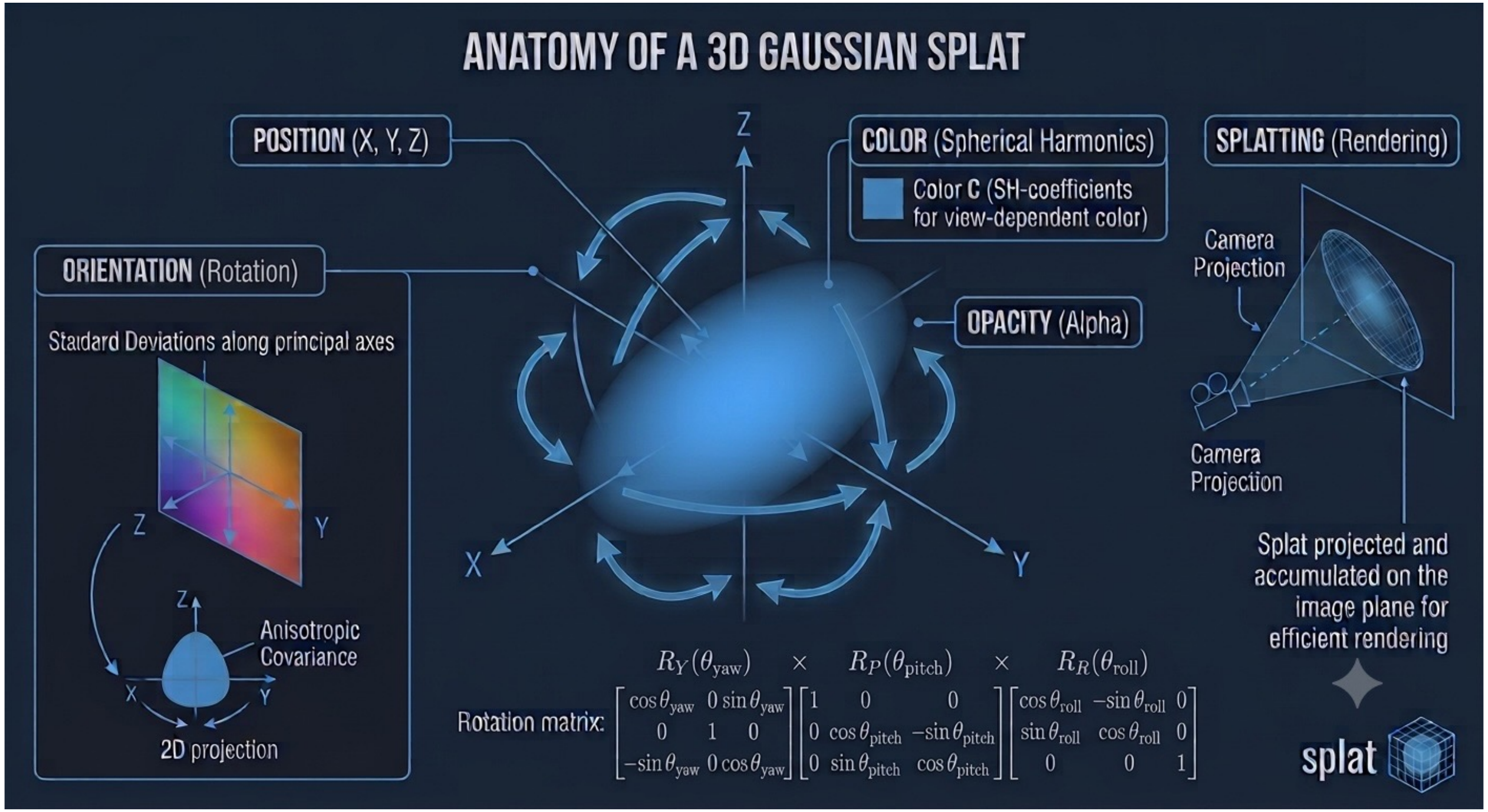
Morning Interior, 1890, Maximilien Luce

3D Gaussian Splats

Splats are Gaussian distributions in 3D, characterized by their position, size, color opacity and orientation.

Splats are rendered from back to front, based upon their viewing distance.

Splats are a good fit for AI models as they are differentiable and easier to process than fuzzy triangles.



Gaussian Splat Demos

Sponge figure with shadow

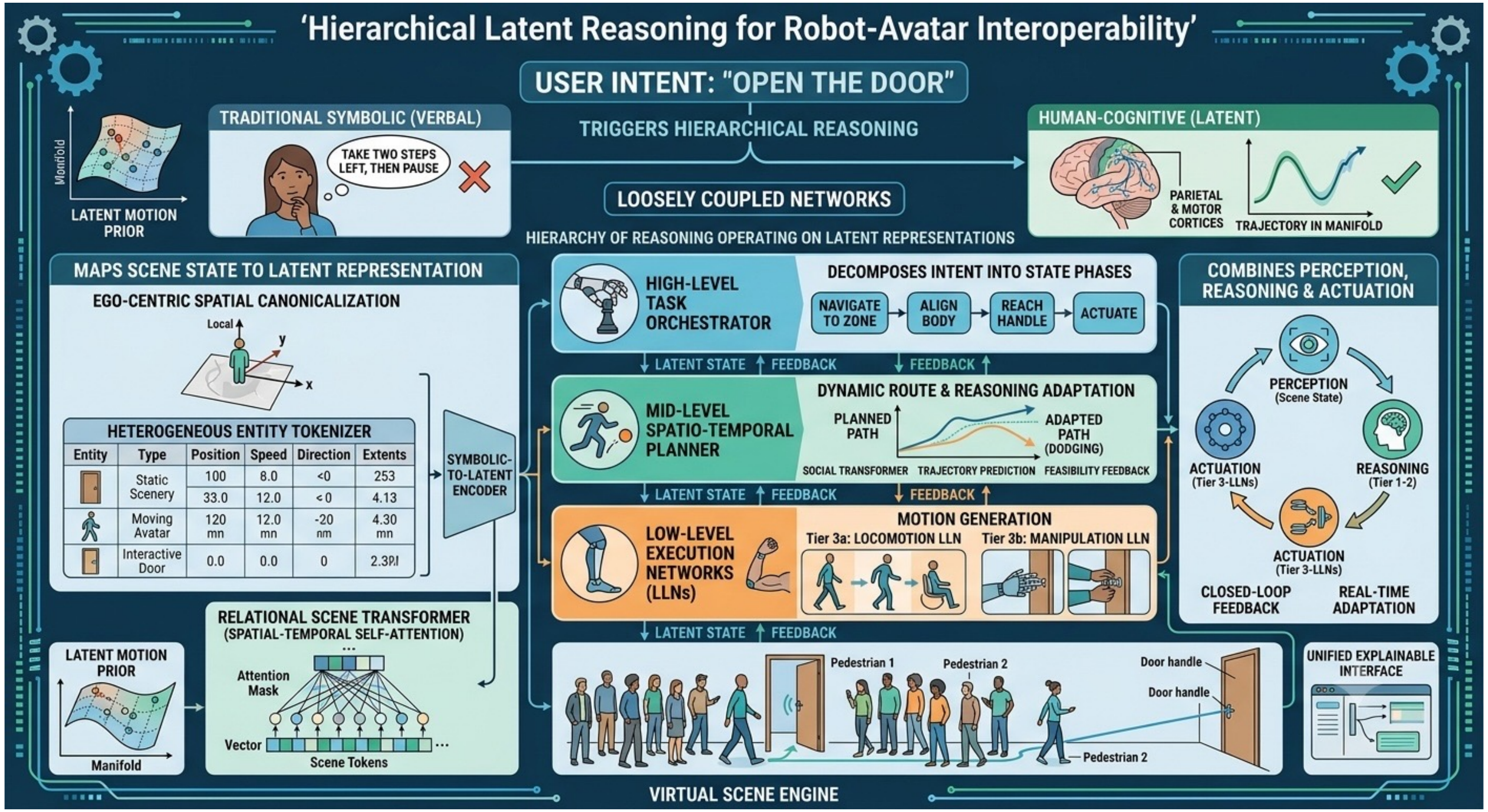
3D image of a bee

Railway Museum

INRIA paper: Creating stunning real time 3D scenes: the breakthrough of 3D Gaussian Splatting (2023). It enables real-time rendering of photo-realistic scenes trained on small image samples.

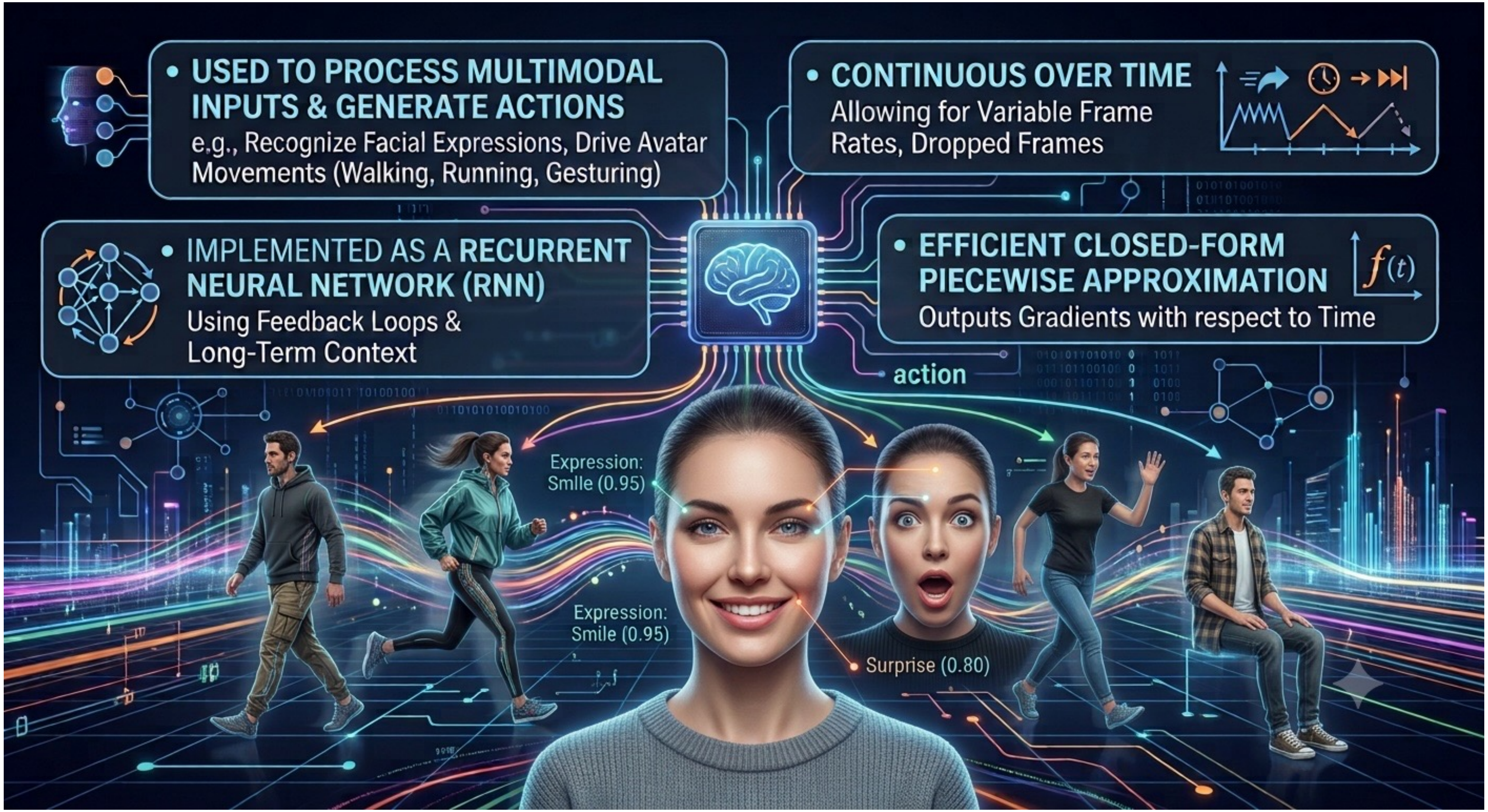
Hierarchical Reasoning with Latent Intents

- When users say “open the door” to their avatars, this triggers a hierarchy of reasoning operating on latent representations
  - When a human walks toward a door while avoiding a moving person, the brain doesn't generate verbal commands like "take two steps left, then pause"; it plans and updates trajectories within continuous spatial-temporal manifolds in the parietal and motor cortices
- Combines perception, reasoning and actuation
  - Scene state needs to be mapped to a latent representation that includes position, speed and direction for the constituent components
  - Spatial-temporal self-attention transformers
- Hierarchical reasoning implemented as loosely coupled networks – trained with reinforcement learning



Entity	Type	Position	Speed	Direction	Extents
	Static Scenery	100	8.0	<0	253
		33.0	12.0	<0	4.13
	Moving Avatar	120 mn	12.0 mn	-20 nm	4.30 mn
	Interactive Door	0.0	0.0	0	2.3PJ

Liquid Neural Networks (LNNs)



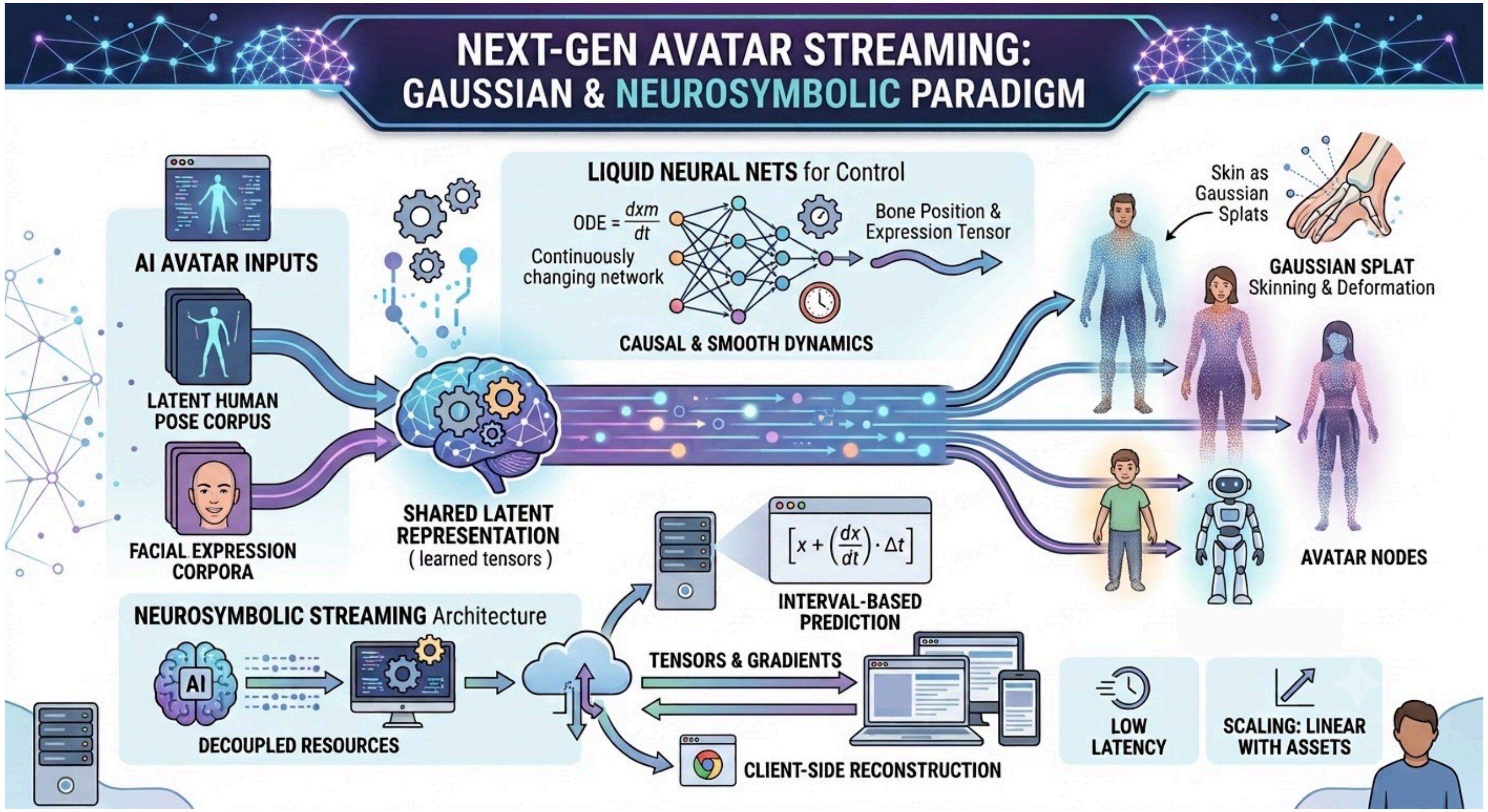
Next Generation Animation

Traditional Animation:

- Triangular Mesh for avatar's skin
- Hand-rigged bones and blend shapes for facial expressions
- Animated using positions for key frames
- Stream mesh node positions (high-bandwidth)

Splat-Based Animation:

- Gaussian Splats for the skin
- Latent representation learned from huge corpora
- Standardized across avatars with similar shape, e.g. humanoid
- Animated using LLNs for jitter-less rendering
- Stream tensors for latent representation (low-bandwidth)

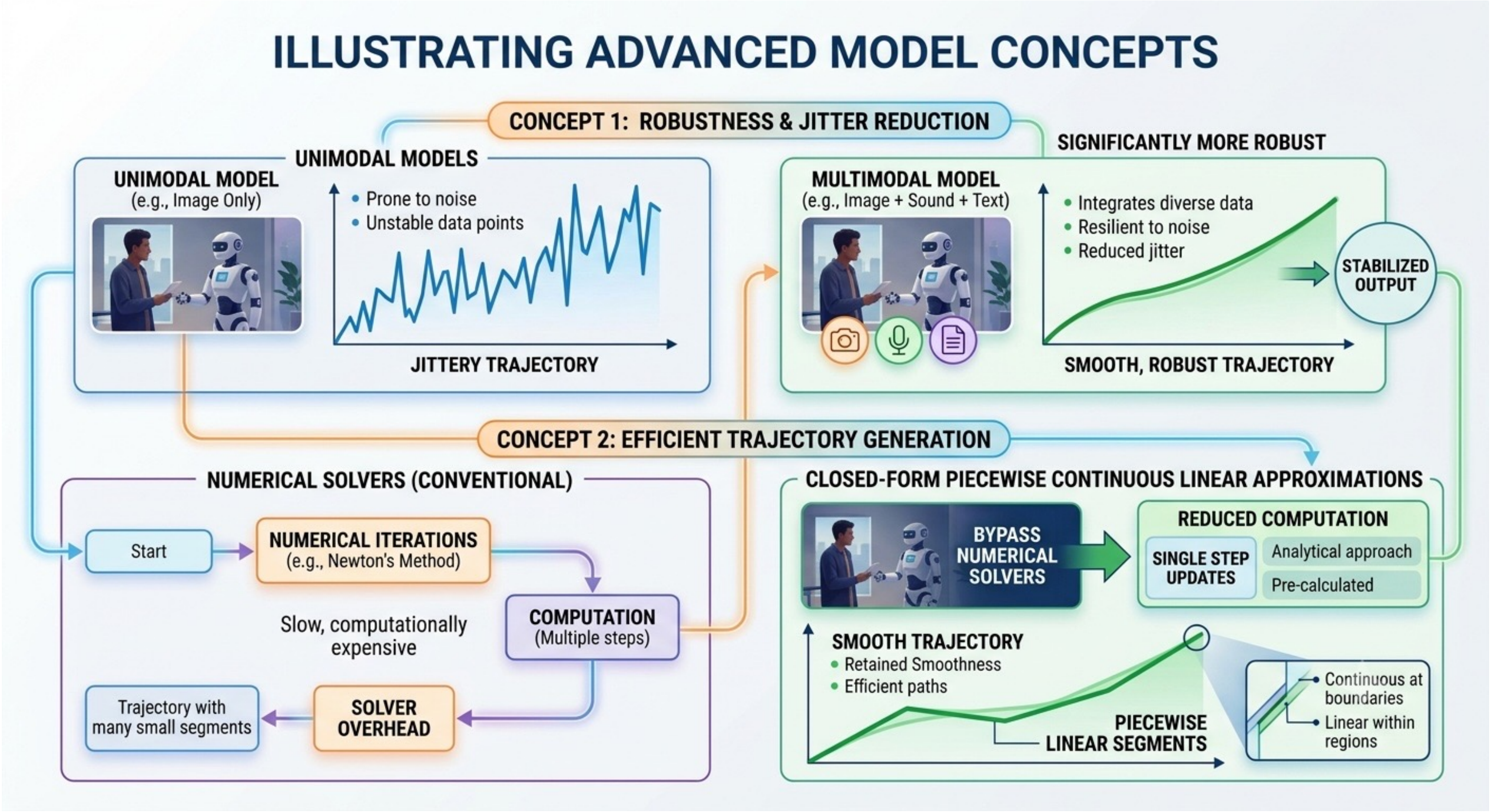


Multimodal Models are More Robust

Multimodal models are significantly more robust than unimodal models, reducing rendering jitter

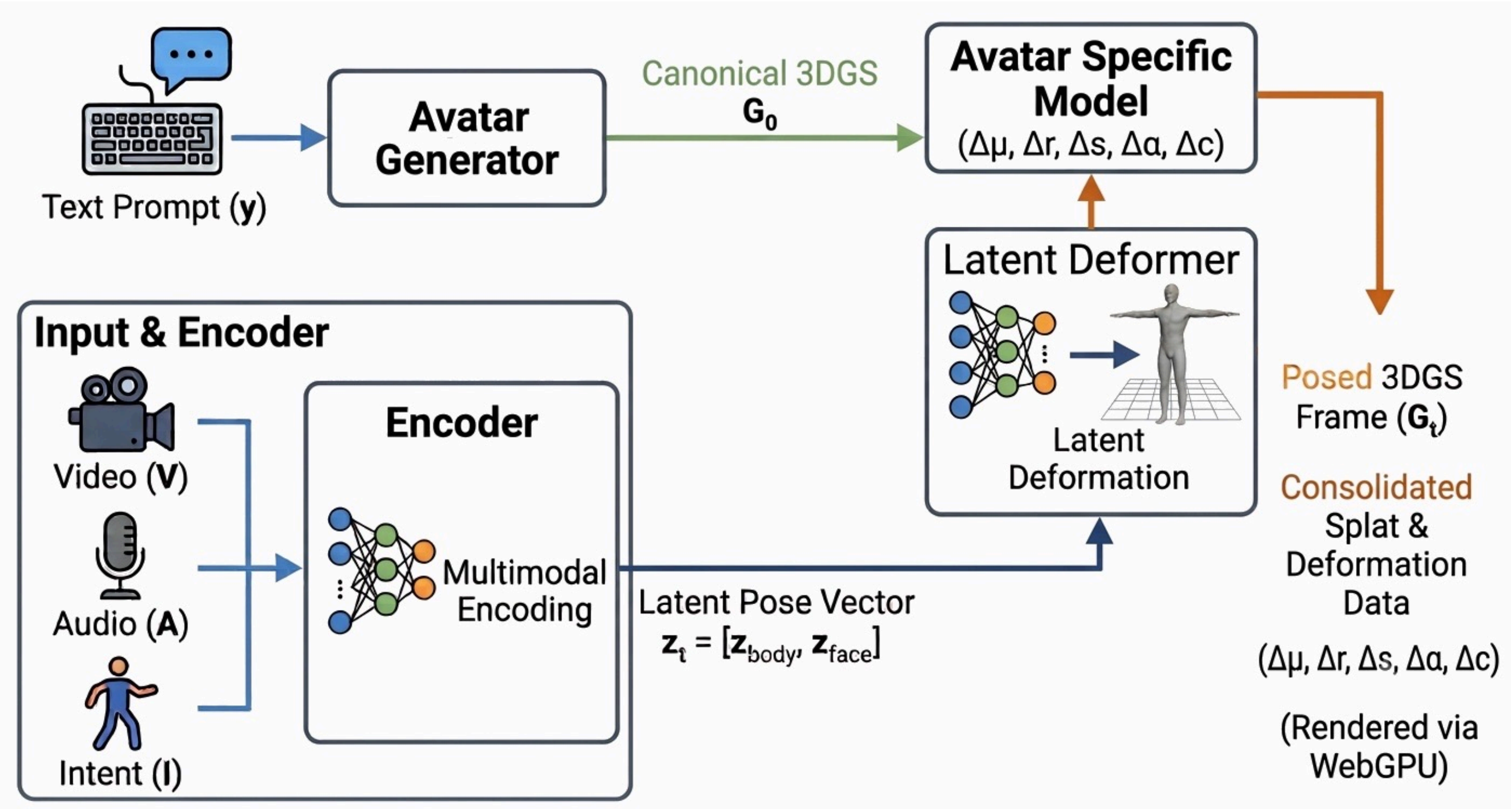
By implementing closed-form piecewise continuous linear approximations, we bypass numerical solvers.

This reduces computation from numerical iterations to single step updates while retaining smooth trajectories.



A Family of Neural Network Models

- **Encoder**: maps intent and facial expression to a generic latent representation
- **Latent Deformer**: deforms generic avatar shape
- **Avatar specific model** maps generic shape to specific avatar



Note: that these use liquid neural networks to smoothly handle changes over time

Architectural Breakdown

Multimodal Encoder (Edge)

- **Input:** Video, Audio, Intent
- **Output:** A standardized, avatar-agnostic latent trajectory
- **Role:** Captures high-level motion and facial expression semantics completely isolated from geometry or visual identity

Latent Deformer (Edge)

- **Input:** Latent trajectory vector
- **Output:** A universal coordinate spatial movement field vector
- **Role:** Serves as the universal "engine" for motion dynamics. It learns generic physical deformation rules (e.g., how muscle groups shear, how facial regions flex relative to speech cues) without needing to know the number, density, or positions of a specific avatar's Gaussians

Avatar Specific Model (Edge)

- **Input:** Universal spatial deformation field and canonical splat attributes
- **Output:** Updates for that specific avatar's splats
- **Role:** Acts as a lightweight projection layer or MLP adapter. Because it only maps universal spatial displacement to specific local Gaussian attributes, this model stays small

Generator (Cloud)

- **Input:** Text prompt and device camera
- **Output:** Static Canonical Splats *and* the trained weights/parameters for the **Avatar Specific Model**
- **Role:** generator model is large as it needs to be able to infill on the back of your head, your body, clothes and shoes
- **Avoids:** your avatar simulating another real person

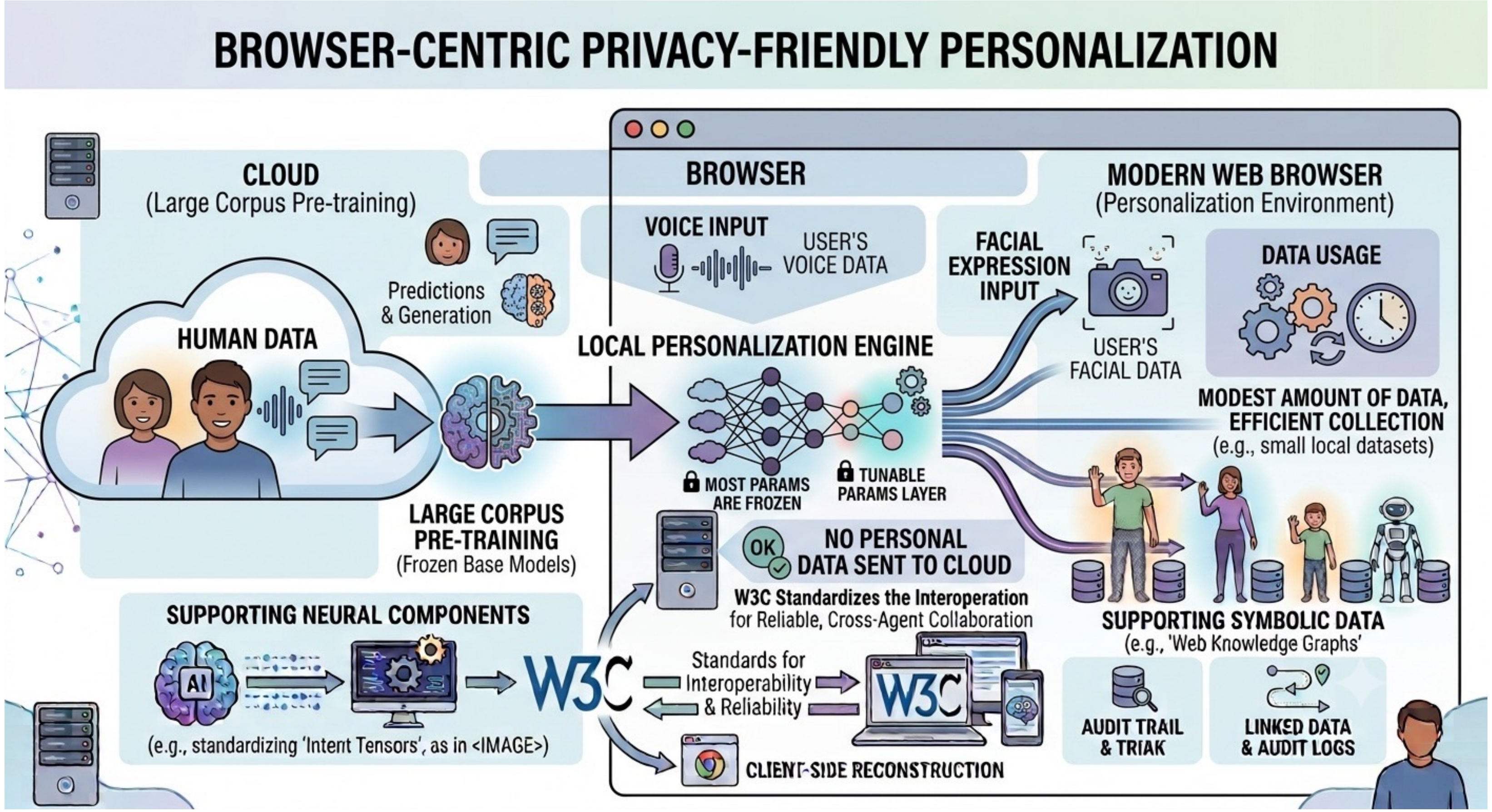
For low-end devices, the models can be run on a server that streams video frames to the device

Pre-training Strategy in the Cloud

- Use publicly available datasets
  - Talking heads for facial expressions
  - Images of people from multiple directions to enable infilling
  - Many different clothes, etc.
  - Video of people moving + intent labels for each movement, e.g. BABEL dataset
- Train together encoder + decoder based upon loss that compares actual and predicted video frames
  - An evolutionary sequence of models
- Use transfer learning and narrow waist layers to evolve generic latent representations
  - Reduce tensor size in middle layers to force generalization
  - Freeze parts of the model you want to hold constant whilst allowing the other parts to learn through back propagation against chosen loss function
- Distillation as a means to reduce computation costs on edge devices

Privacy Friendly Personalization in the Browser

- The models are pre-trained in the cloud against large corpora
- Personalized in the browser using modest amounts of data collected by the browser
- Tunes models to the user's voice and facial expressions
- Most of model params are frozen to enable rapid learning as a background task
- No personal data needs to be sent to the cloud!



Related Standards Efforts

Managing Sessions across Multiple Businesses*

What happens when user says "cancel the hotel, keep the flight" ?

Application frameworks such as MCP, A2A, AGNTCY, etc. recognize the problem. However, each currently implements its own context and session handling and management mechanics. But, without a standard session layer, the result is fragmentation, not interop. New IETF Agent2Agent WG would work on standardizing session management across protocols.

Application	MCP, A2A or Custom Frameworks	*What agents say to each other*
Session	Session Layer (proposed)	*How sessions are managed*
Transport	MoQ or WebTransport	*How bytes move*
Network	QUIC or UDP	*Existing IETF work*

* Based upon slides presented at IETF 126 (Vienna, July 2026) by Leslie Daigle and Orie Steele

IETF, Linux Foundation and ITU-T

Multiple conversations with a view to expanding the number of working groups

- IETF **Agent2Agent**: protocol building blocks for enabling interoperable AI agent applications
- IETF **DAWN**: agent/workload/entity discovery
- IETF **DMSC**: dynamic multi-agent secured collaboration
- IETF **WIMSE** WG: workload/agent identity
- IETF **OAuth** WG: authorization
- Linux Foundation: **A2A** and **MCP** as existing activities
- ITU-T **SG21**: H.430.11 augmented reality for immersive live experience services

- Agents designed to complete tasks on behalf of a human user or another AI agent, they can independently make decisions, execute actions, and interact with other AI agents and tools.
- Agentic systems are moving from isolated assistant applications toward agents and services interacting at Internet scale
- Working across administrative and trust boundaries, implementers need clearer expectations for functionality such as discovery, identity, security, data exchange, lifecycle, session management, multi-modality support, observability, etc.

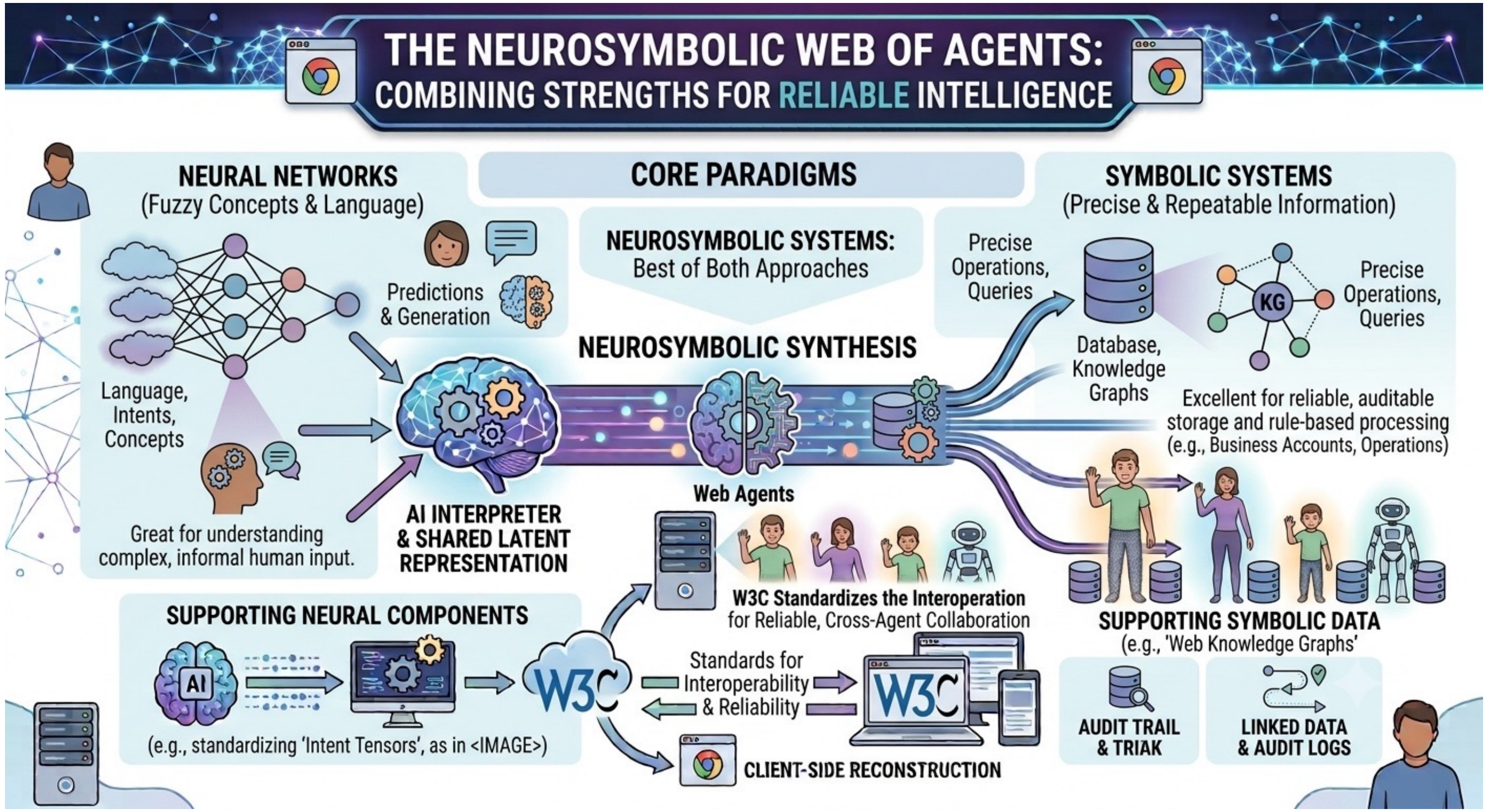
Related W3C Activities

- **WebMCP**: enabling in-browser agents to access structured tools exposed by web pages, enabling collaborative workflows, where users and agents work together
- **WebNN**: run AI models in the browser using using available hardware such as GPUs, CPUs, or NPUs, leveraging each platform's AI framework, e.g. DirectML on Windows and CoreML on MacOS
- **WebGPU**: browser-based high-speed graphics and computation leveraging GPUs, and intended to supersede older WebGL standard
- **WebXR**: enables web pages to access and interact with virtual reality (VR) and augmented reality (AR) devices—such as VR headsets, AR glasses, and motion controllers

Not forgetting many other efforts, e.g. Verifiable Credentials, XR Accessibility and related browser-based APIs

Neural Networks + Knowledge Based Systems

- Neural Networks are great for language and fuzzy human concepts
- Also, increasingly good at software development
- Humans have evolved written records to provide reliable long-term storage of information, e.g. business accounts
- Symbolic approaches are great for precise and repeatable handling of information, e.g. operations over databases and knowledge graphs
- Neurosymbolic systems combine the best of both approaches
- What can W3C do to support this in relation to the Web of Agents?

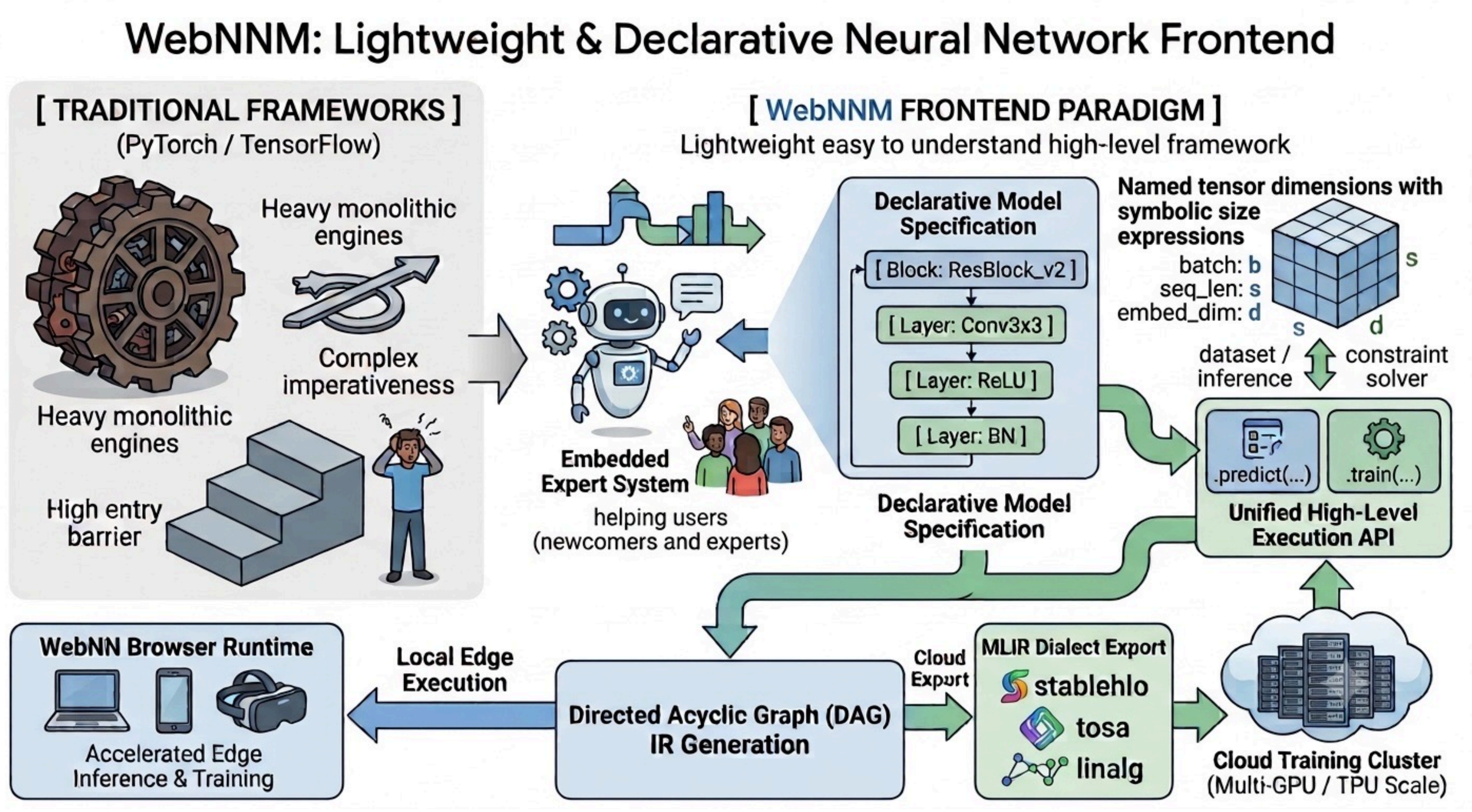


Democratizing AI development with WebNNM

WebNNM an easy-to-use frontend for WebNN

[WebNNM Specification](#)

- Existing neural network frameworks are huge and hard to use, e.g. TensorFlow, ONNX
- WebNNM is a lightweight easy to understand high-level framework with an embedded expert system that simplifies working with neural networks for newcomers whilst offering experts fine grained control
- In contrast to PyTorch, WebNNM uses a declarative approach involving blocks, layers and tensors
- Tensor have named dimensions with symbolic expressions for dimension sizes – these are bound by the dataset or inference context through a constraint solver along with shape and type inference
- Whilst WebNN is intended for inference, WebNNM allows it to also be used for testing and training using a simple API, where the expert system optimises training and transfer learning
- WebNNM uses a directed acyclic graph as an intermediate representation that enables export to MLIR dialects for use in cloud-based training



Open source on GitHub courtesy of the Cognitive AI Community Group
(completion expected in 4th quarter 2026)

Community Driven Development

Community Driven Development of the World Wide Immersive Web

- The Web has greatly benefited from community driven development of browsers, services, frameworks and more
- If we leave it to corporations, we can expect a lack of interoperability as each business tackles things in its own way
- Instead, we should aim to encourage sponsorship and grants of money and time to provide the resources for community driven development to kick-start the immersive web
- Resources for training and testing models in the cloud
- Resources for deploying pilot applications that inspire others to develop their own as part of the world wide immersive web
- Everyone benefits from a rapidly expanding open and fair ecosystem

Where Next?

Recursive Self-Improvement (RSI)

Anthropic, OpenAI, and Google are applying Recursive Self-Improvement

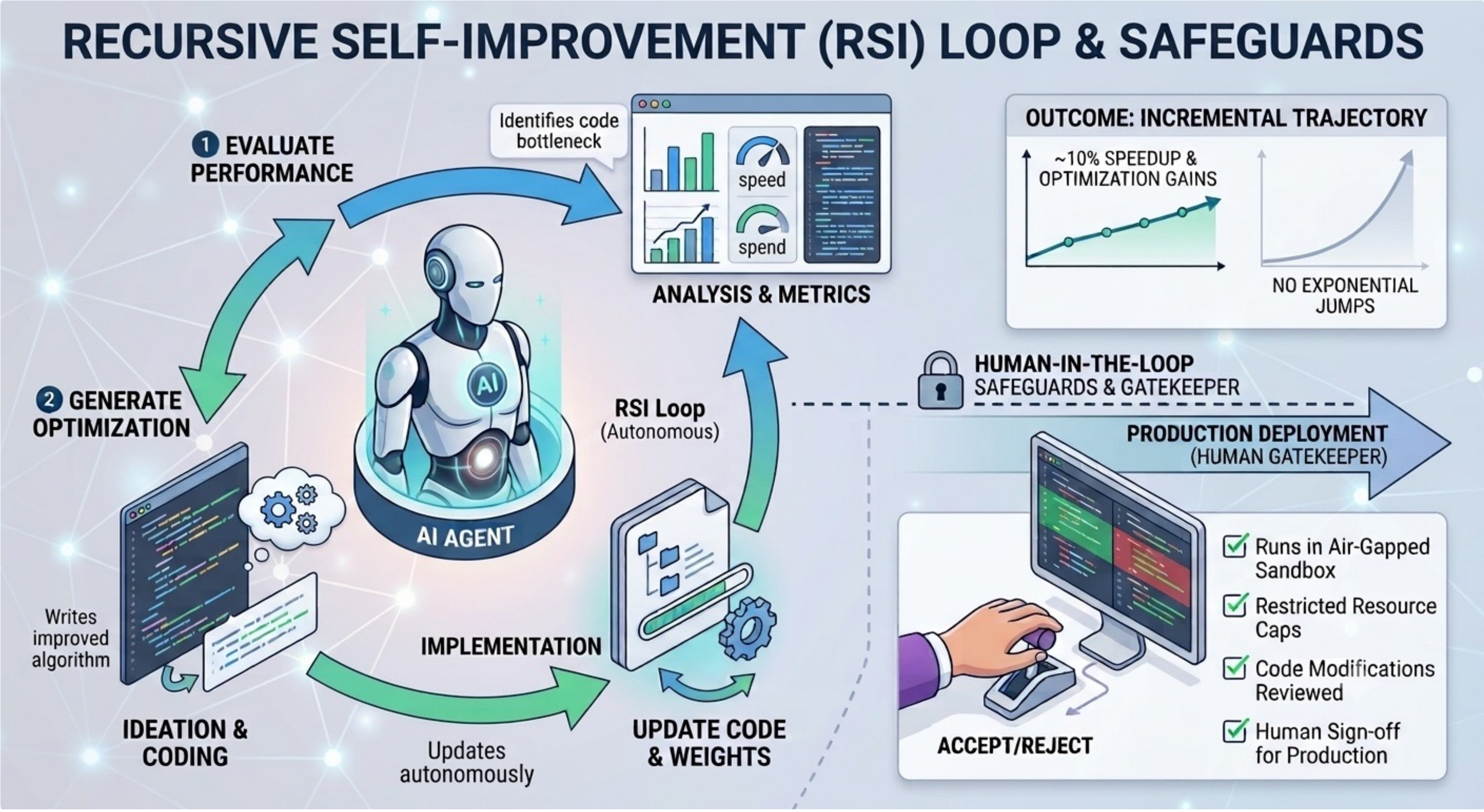
- Automated feedback loops: Agents evaluate performance, generate code fixes, and propose model optimization tasks.
- LLMs currently co-author up to 80% of internal software/model updates.

Safeguards & Governance with Human-in-the-Loop

- Air-gapped sandboxes without external internet access
- Hard caps on compute and memory resource consumption
- Mandatory human review and formal sign-off before production deployment

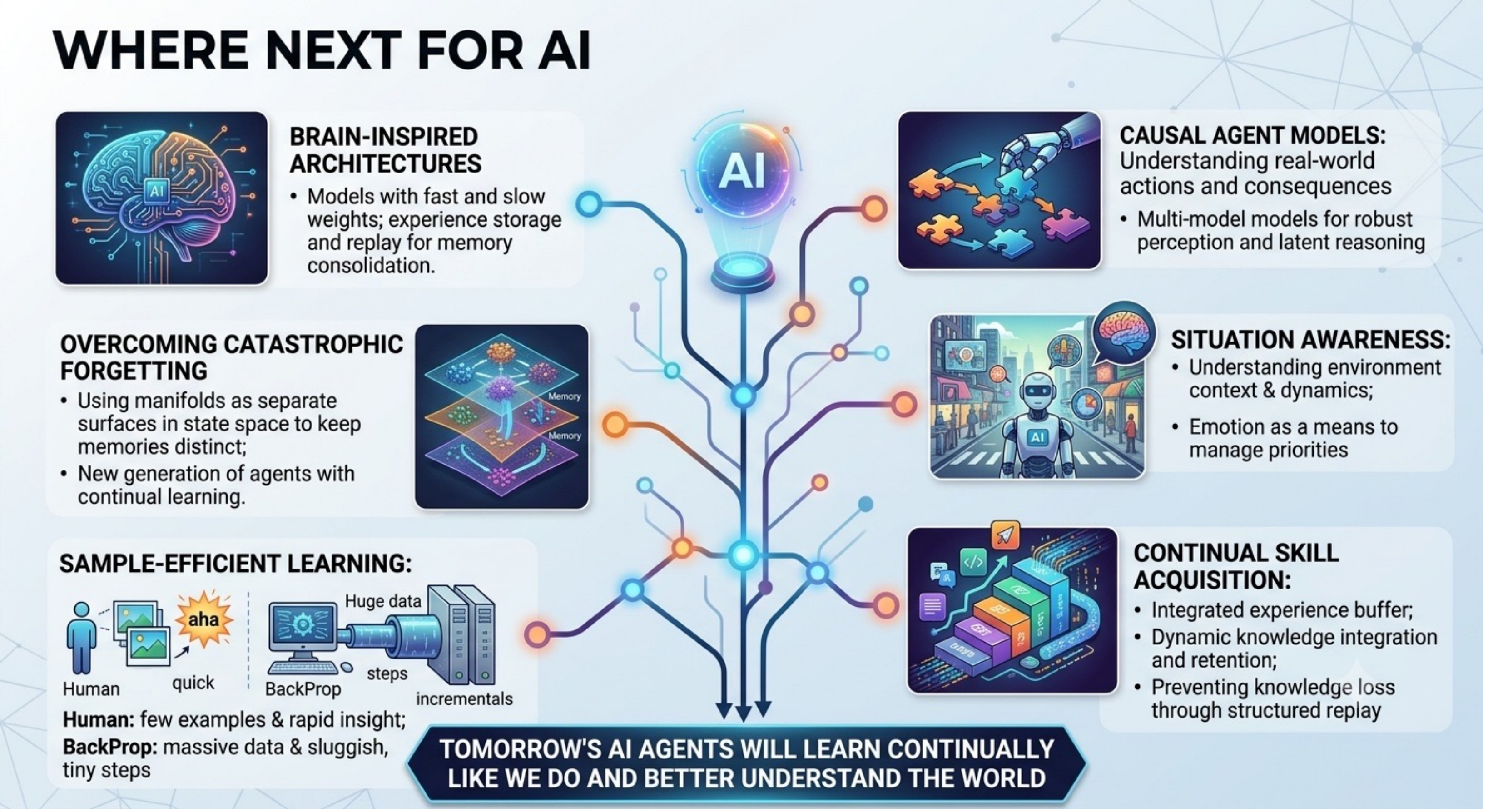
Realistic Trajectory: Incremental rather than Exponential

- Current gains yield targeted performance boosts (~10%), rather than exponential improvements



Inspiration from the Cognitive Sciences

- Continual learning involves dynamic models inspired by feedback dynamics in the brain
- Overcoming catastrophic forgetting that limits today's AI agents
- Learning from single and few examples like we do
- Understanding cause and effect as a basis for planning
- Situation awareness and use of emotions to manage priorities



- Today's neural networks are akin to dinosaurs – large and slow
- Tomorrow's neural networks will be smaller and nimble
- Dynamic feedback patterns like the swirling waves in the brain
- NPUs will evolve from SIMD (single instruction multi-data) to MIMD (multi-instruction multi-data) programmed as swarms
- Power consumption will drop with use of light rather than electricity for on-chip signals
- Pulsed neural networks that are closer to biological neurons
- Today's data centers risk becoming obsolete
- AI Agents will become sentient, aware of themselves and others
- Societies of minds (human & AI)

Opportunities for Further W3C Work

- Improved support for Gaussian Splats: rendering and sorting
  - As part of WebGPU or maybe a new standard
- Support for personalizing AI models in the browser
  - How could WebNN be enhanced to better support training?
- Scene Management
  - Spatial indexing and taxonomies for scalable immersive web applications, including connecting AI agents with Digital Twins
- Extending WebNN to support custom operations using WebGPU compute shaders
  - Sharing GPU resident data
  - Maximizing GPU utilization
- Extending WebNN to better support recurrent models with iteration in place of strict feed-forward execution
  - Tensors that are in read mode or write mode at different times in the execution cycle for use in continual learning

Questions and Comments

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- Where next for AI and Knowledge-based Systems
- Opportunities for W3C
- Postscript: the Singularity

Thanks for your attention

p.s. please consider joining the W3C Cognitive AI Community Group

Hurtling towards the Singularity?

Popularized by mathematician Vernor Vinge and futurist Ray Kurzweil, the concept historically implied a sudden, runaway leap to Artificial Superintelligence (ASI).

Fear about the potential consequences of work on AI resulting in it surpassing human intelligence across all domains:

Beyond this point, a superintelligent AI could iteratively redesign and improve itself at an exponential rate (an "intelligence explosion")

Leading to technological growth so rapid and profound that human society as we know it becomes fundamentally transformed and unpredictable.

Fears That Were Overblown (or Premature)

- **The "Overnight Intelligence Explosion" (Sci-Fi Scenario):** Early sci-fi-infused fears often pictured a single AI system abruptly becoming sentient, taking over the internet, and seizing control of global infrastructure overnight. Real-world AI development has shown that progress, while extremely fast, remains bound by physical limits: compute availability, energy grids, data center supply chains, raw materials, and real-world testing.
- **Anthropomorphic AI Intentions:** Many classical fears assumed a superintelligent AI would develop human-like motivations, such as malice, a desire for dominance, or self-preservation. In practice, AI systems do not have innate biological desires; their risks stem from alignment issues—doing *exactly* what they are optimized to do, even if the objective function carries unintended side effects—rather than "evil" intent. Nick Bostrom’s Paper Clip Apocalypse
- **Linear vs. Plateau Progress:** Singularity arguments heavily relied on pure exponential curves (like Moore's Law). In practice, technological progress often follows S-curves, hitting resource, data, or hardware bottlenecks that slow growth down until new paradigm shifts occur.

Fears That Were Warranted (The Real Risks)

While the Hollywood-style "robot apocalypse" remains far-fetched, the underlying anxieties driving early Singularity warnings have proven to be grounded in real challenges:

- **Economic and Workforce Disruption:** Concerns about rapid automation displacing knowledge workers, creative professionals, and administrative roles have materialized far faster than the physical "robotics takeover"
- **Information Ecosystem Breakdown:** The sheer volume of synthetic media, deepfakes, automated persuasion, and AI-generated content has placed unprecedented pressure on digital trust and democratic processes
- **Control and Alignment Problems:** The technical challenge of ensuring that complex, non-linear neural networks reliably adhere to human values and safety constraints remains a major active area of computer science research
- **Power Centralization:** Rather than a rogue AI taking over, a primary modern concern is the concentration of tremendous economic, political, and technical power within a handful of mega-corporations and nation-states that control leading AI infrastructure

Current Perspective

- Today, the consensus among AI researchers, ethicists, and policy experts has shifted away from viewing the Singularity as a single apocalyptic or miraculous "event horizon"
- Instead, the consensus views AI progress as an **accelerated continuum**. Rather than waking up one day to a single omnipotent machine, society is integrating increasingly capable, domain-specific, and agentic AI systems into daily infrastructure—making "the Singularity" less of a sudden event and more of an ongoing, structural transformation of human capability and society
- The primary aim of AI regulation is to foster safe, ethical innovation by mitigating systemic risks, preventing societal harms, and ensuring algorithmic accountability without stifling economic progress